Vector Databases and Vector Search

1. Vector Search

Vector search is a retrieval technique that compares a query vector with vectors stored for documents or document chunks. It is useful when relevance depends on semantic similarity rather than exact text matching.

2. Indexing

During ingestion, documents are split into chunks and converted into embeddings. The resulting vectors and associated metadata are stored in a vector index or vector database. Metadata can include document identifiers, titles, pages, or other source information.

3. Querying

At query time, the user's question is converted into an embedding. The search system compares that vector with stored vectors and returns the highest-ranked matches. These matches become candidate context for downstream processing.

4. FAISS

FAISS is a library for efficient similarity search over dense vectors. It can be used to build an in-memory or persisted vector index for applications that need fast nearest-neighbor retrieval.

5. Vector Databases

Vector databases provide storage and retrieval capabilities for vector data, often together with metadata filtering and other database features. Examples of technologies used for vector retrieval include pgvector with PostgreSQL and dedicated vector database systems.

6. Limitations

Vector search can retrieve semantically related text that is not actually sufficient to answer a question. It can also miss exact terms, identifiers, names, or rare keywords. For this reason, many RAG systems combine vector search with keyword retrieval.